

Digital health in the Eastern Mediterranean Region

A snapshot





Digital health in the Eastern Mediterranean Region

A snapshot



**World Health
Organization**

Eastern Mediterranean Region

Digital health in the Eastern Mediterranean Region. A snapshot.

ISBN 978-92-9274-360-4 (electronic version)

ISBN 978-92-9274-359-8 (print version)

© **World Health Organization 2024**

Some rights reserved. This work is available under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 IGO licence (CC BY-NC-SA 3.0 IGO; <https://creativecommons.org/licenses/by-nc-sa/3.0/igo>).

Under the terms of this licence, you may copy, redistribute and adapt the work for non-commercial purposes, provided the work is appropriately cited. In any use of this work, there should be no suggestion that WHO endorses any specific organization, products or services. The use of the WHO logo is not permitted. If you adapt the work, then you must license your work under the same or equivalent Creative Commons licence. If you create a translation of this work, you should add the following disclaimer along with the suggested citation: "This translation was not created by the World Health Organization (WHO). WHO is not responsible for the content or accuracy of this translation. The original English edition shall be the binding and authentic edition".

Any mediation relating to disputes arising under the licence shall be conducted in accordance with the mediation rules of the World Intellectual Property Organization.

Suggested citation. Digital health in the Eastern Mediterranean Region. A snapshot. Cairo: WHO Regional Office for the Eastern Mediterranean; 2024. Licence: CC BYNC-SA 3.0 IGO.

Cataloguing-in-Publication (CIP) data. CIP data are available at <https://iris.who.int/>.

Sales, rights and licensing. To purchase WHO publications, see <http://apps.who.int/bookorders>. To submit requests for commercial use and queries on rights and licensing, see <http://www.who.int/about/licensing>.

Third-party materials. If you wish to reuse material from this work that is attributed to a third party, such as tables, figures or images, it is your responsibility to determine whether permission is needed for that reuse and to obtain permission from the copyright holder. The risk of claims resulting from infringement of any third-party-owned component in the work rests solely with the user.

General disclaimers. The designations employed and the presentation of the material in this publication do not imply the expression of any opinion whatsoever on the part of the World Health Organization concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted lines on maps represent approximate border lines for which there may not yet be full agreement.

The mention of specific companies or of certain manufacturers' products does not imply that they are endorsed or recommended by the World Health Organization in preference to others of a similar nature that are not mentioned. Errors and omissions excepted, the names of proprietary products are distinguished by initial capital letters.

All reasonable precautions have been taken by the World Health Organization to verify the information contained in this publication. However, the published material is being distributed without warranty of any kind, either expressed or implied. The responsibility for the interpretation and use of the material lies with the reader. In no event shall the World Health Organization be liable for damages arising from its use.



Contents

Foreword	v
Acknowledgements.....	vi
Abbreviations.....	viii
Executive summary	ix
Introduction	1
Results	5
Survey responses	5
1. Health priorities	5
2. Digital transformation commitment (policies and strategies)	5
3. Legislation for digital applications	5
4. Capacity-building for health workers on the use of ICT for health	5
Discussion	15
Foundations of digital health	15
Digital health services and applications	15
Digital health standards and interoperability	16
Perceived barriers to adopting and implementing digital health programmes	16
Recommendations	17
Conclusion	18
References	19
Annex 1. Digital health country profiles	20
Annex 2. Digital health national strategy cross-check using survey response and publicly available data	43
Annex 3. Electronic health records cross-check using survey response and publicly available data	45
Annex 4. Reported national health priorities	47
Annex 5. WHO regional survey tool	51



Foreword

WHO's global vision of digital health is to improve health for everyone, everywhere by accelerating the development and adoption of appropriate digital health solutions to prevent, detect and respond to epidemics and pandemics, developing infrastructure and applications that enable countries to use health data to promote health and well-being, and to achieve the health-related Sustainable Development Goals and the Triple Billion targets of WHO's Thirteenth General Programme of Work, 2019–2023.

The COVID-19 pandemic highlighted the vital importance of opportunities presented through digital health, as well as a realization of the unique barriers to digitalization of the health sector, which include complicated workflows, fragmented systems and tools, a lack of standards and interoperability, privacy and security risks and weak governance in digital health in the Region.

Despite noticeable progress in the Region in the past few years, many countries still lack proper digital ecosystems to reach the right people, at the right time and in the right place.

This regional survey provides an opportunity to explore the challenges related to digital health implementation at country level and how to use this information to analyse gaps in health care that can be reduced through digitalization.

The survey provides a foundation for further action to be taken to support health or digital health in the Region. Understanding the situation (diagnosis) will help in suggesting solutions that will help to identify the needs of each Member State individually and for the Region collectively in monitoring and coordinating digital health investments.

Arash Rashidian,
Director of Science, Information and Dissemination Department,
WHO Regional Office for the Eastern Mediterranean

Acknowledgements

The World Health Organization gratefully acknowledges the contributions of many individuals who have provided inputs to the development of the regional digital health survey and helped to ensure its completion through support from Member States of the Eastern Mediterranean Region.

The survey report was prepared under the leadership of Arash Rashidian, Director of Science, Information and Dissemination Division, WHO Regional Office for the Eastern Mediterranean, with the support of the Digital Health and Innovation team Ahmed Mandil, Mohamed Nour and Ahmed El-Arousy, with further input from WHO consultants Saif Khairat and Mostafa Monir, and WHO intern Ali Bakhsh Nasrat.

Special thanks go to WHO country offices and national focal points in countries/territories of the Eastern Mediterranean Region for their cooperation in completing the regional digital health survey.

Country	Country office respondent(s)	National respondent(s)	National entity
Afghanistan	Najibullah Safi/Saha Naseri	Ahmad Nawid Shams	Ministry of Public Health
Bahrain	Nadia Alaali	Shafeea Alnattei	Ministry of Health
Djibouti	Mohamed Elhakim	Omar Moussa Ali	INSTAD Statistics Department of Djibouti
Egypt	Randa Abou El Naga/ Gasser Gad El Kareem	Amany Gayed	Ministry of Public Health and Population
Iran (Islamic Republic of)	Shadrokh Sirous MashyaneHaddadi	Mahmood Tara/ Ali Sharifi-Zarchi	Minister's Directorate for Statistics and ICT
Iraq	Aamr Bebany	Ali Hussein Hassan Al Temimi/ Alaa Hussein Ali Alameri	Ministry of Health
Jordan	Novera Ansari	Feras Kamal/ Fatima Hammad	Hakeem/ Ministry of Health
Kuwait	–	Dari Alhuwail	Kuwait University, Department of Information Science, and a member of the WHO's Digital Health Global Advisory Group
Lebanon	Alissar Rady	Linal BouMrad	e-Health unit at the Ministry of Public Health
Libya	Haroun El Rashid/ Mohamed Hashem	Mohammad Ibrahim Daganee	Director, Health Information Centre, Ministry of Health
Morocco	Hachri Hafid	Abdelkader Mhayi	Ministry of Health
Oman	–	Halima Al-Hinai	Ministry of Health
Pakistan	Shazia Ishaque/ Syed Baqar Hasnain Jafri	Hasan Bin Hamza	Ministry of Health, Technical Working Group on eHealth
Palestine	Abu El Hawa, Munia	Ali Helou	Ministry of Health
Qatar	–	Juliet Ibrahim	National eHealth Committee
Saudi Arabia	Khaled Nada	Abdulaziz Abdullah Alrumaih	Ministry of Health
Somalia	Marina Madeo	Hassan Sh Ahmed	Ministry of Health, Federal Government of Somalia

(continued)

Country	Country office respondent(s)	National respondent(s)	National entity
Sudan	Muhyeldeen Abdalla Mohammadi Salih	Mahmoud Abdelhadi	Ministry of Health
Syrian Arab Republic	Hussam Al Halak	Azza Abdrabo	Ministry of Health, IT department
Tunisia	Ramzi Ouhichi/ Priyanka Saksena	Faouzi Mehdi/ Lotfi Allani	IT Centre of the Ministry of Public Health)
United Arab Emirates	–	Mubaraka M. Ibrahim / Ali Juma Khamis Alajme	Ministry of Health and Prevention
Yemen	Muaadh Thabit/ Manhal Al Halasa	–	–

Abbreviations

GOe	Global Observatory for e-Health
HIS	Health information system
ICT	Information and communication technology
ITU	International Telecommunication Union
PAHO	Pan American Health Organization
SDGs	Sustainable Development Goals
UHC	Universal health coverage
WHA	World Health Assembly
WHO	World Health Organization



Executive summary

Digital health has been proven to enhance health care delivery and improve the accessibility, quality, efficiency and cost-effectiveness of national health systems. The World Health Organization (WHO) has approved several resolutions to advance digital health since 2005. The latest in the series is the *Regional strategy for fostering digital health in the Eastern Mediterranean Region (2023–2027)*, endorsed at the 69th session of the Regional Committee in October 2022.¹ This is founded on the *Global Strategy on Digital Health 2020–2025*, published in 2021² following a resolution of the World Health Assembly in May 2018.³

Digital health assists countries in attaining universal health coverage (UHC), especially in rural, remote and migrant settings and communities, as well as in enhancing emergency response and monitoring of processes and outcomes. The opportunities and challenges of digital health were particularly noticeable during the COVID-19 pandemic response. Digital health offered opportunities for rapidly implementing digital health applications and solutions. At the same time; however, weak and fragmented health data governance, lack of data security and privacy and lack of protection against risks from improper use of data were reported as some of the key challenges during this period.

WHO conducted two major regional surveys in 2017–2018 and 2021–2022 across the 22 countries/territories of the Eastern Mediterranean Region. The aim of the 2017–2018 survey was to review and assess the status of digital health in countries of the Region based on six themes: national health priorities; national health information policies or strategies; services and applications; standards and interoperability; governance and guidance; and perceived barriers to the adoption and implementation of digital health in the country.

The 2021–2022 survey version was developed in Excel embedding the responses of the 2017–2018. It aimed to review and validate responses and contained an additional section to assess digital health interventions and perceived challenges during the COVID-19 pandemic. Responses to the questions asked by the most recent survey gave the following results.

Foundation for digital health

Health priorities: A wide range of health priorities were identified by countries, ranging from disease surveillance management, health information strengthening and broader areas relating to achievement of UHC. The identification of health priorities form the basis of stakeholders' dialogue in developing strategic plans. The reported priorities are contained in Annex 3.

Digital transformation commitment (policies and strategies): Commitment to digital transformation was ascertained through identifying the existence of policies and strategies relating to the use of ICT or e-Health, including:

1. National health policies and strategies: Fourteen countries (63.6%) of the 22 in the Region have developed a national health policy or strategy that clearly refers to the use of ICT or e-Health indicating a general commitment to digital transformation and nine countries (40.9%) have developed a strategy or policy to address UHC that refer to ICT or e-Health. Twelve countries (54.5%) have developed, and eight countries (36.4%) are in the process of developing a national health information system policy or strategy. This demonstrates the commitment of about 91% of countries to such policies/strategies. Seven countries (31.8%) have developed, and eight countries (36.4%) are in the process of developing a national e-Health (or digital health) policy or strategy.

¹ Regional strategy for fostering digital health in the Eastern Mediterranean Region (2023–2027) (EMRC69/R.6). WHO Regional Committee for the Eastern Mediterranean; 2022 (<https://applications.emro.who.int/docs/Digital-Health-EMR-2023-2027-eng.pdf>).

² Global strategy on digital health 2020–2025 (WHA73/28). Geneva: World Health Organization; 2021 (<https://www.who.int/docs/default-source/documents/gs4dhdaa2a9f352b0445bafbc79ca799dce4d.pdf>).

³ Digital health draft resolution (WHA71.7). WHO: Seventy-First World Health Assembly, 21 May 2018 (https://apps.who.int/gb/ebwha/pdf_files/WHA71/A71_ACONF1-en.pdf).

2. Existence of legislation to protect the privacy of personally identifiable data of individuals is an indicator of the level of maturity of digital health in a country. Fifteen countries (68.2%) confirmed having such legislation and four (18.2%) are in the process of developing it.
3. In-service training for capacity-building of health professionals is the major factor for successful implementation and utilization of digital health. Sixteen countries (73.7%) reported investing in the continuing development of the existing health care workforce, reflecting recognition of the importance of a digitally competent workforces.
4. Digital health services and applications: The survey investigated the existence of 27 digital health applications classified under six categories. The top application in terms of number of adopting countries was digital surveillance under the category of health care management and administration, followed by electronic medical records under the category of individual electronic health information, then telemedicine and m-Health under the category of health information and knowledge.
5. Digital health standards, interoperability and legal frameworks: Three components of standardization were verified with 68% of countries reported adoption of at least one component. Ten (55.5%) of the countries surveyed have common digital health terminologies, but only 38.8% have data structure standards and 27.7% have software accreditation.
6. Digital health governance and guidance: Almost all countries have defined designated governance and guidance authorities. Fourteen countries are governed and guided by public organizations mostly lead by the ministry of health, in collaboration with the ministry of communication and information technology. Two countries are governed by public-private partnerships and two are governed by WHO and nongovernmental organizations.
7. Perceived barriers to adopting digital health in the Region included: lack of trained human resources, lack of funding for development and support, competing health system priorities, lack of equipment and/or connectivity and lack of legislation or regulations covering e-Health programmes. Most of the barriers stem from lack of qualified leadership who could advocate for the value of digital health and coordinate its implementation and investment decisions.

The results showed that countries had the following services, in ranking order:

- vaccine administration and management system (68%)
- telemedicine/telehealth services (37%)
- diagnosis applications (27%)
- digital contact tracing (27%)
- chatbots (18%)
- mobile health (mhealth) (14%)
- hospital bed dashboard (5%)
- traveller contact system (5%)
- COVID-19 outbreak monitoring platform (5%).

The report concludes that countries in the Region need to develop a roadmap for digital health, strengthen governance through policy and coordination, and leverage the opportunities of advanced digital health solutions to improve the health and well-being of their populations.



Introduction

The use of digital technologies in health is transforming health system access and health service delivery around the globe. Although digital transformation began several decades ago, its real value was clearly demonstrated by the shortfalls exposed in many health systems by the COVID-19 pandemic in 2020 and the role and application of digital solutions in pursuing evidence-based decision-making. Strategies for adopting digital technologies were developed rapidly during the pandemic to ensure continuity of access to high-quality health services, especially for people living with noncommunicable diseases (1). In the Eastern Mediterranean Region, many digital innovations were scaled up in response to the pressing need to improve the accessibility and agility of health services (2). However, without any strategic direction, the adoption of digital health solutions resulted in fragmented, short-term projects with little impact on people's health and well-being. Thus, strategic planning for the appropriate adoption of digital health is key for countries to accelerate progress towards achieving universal health coverage (UHC), the Sustainable Development Goals (SDGs) and the objectives of *WHO's Thirteenth General Programme of Work (GPW13)* (3).

The World Health Organization (WHO) has promoted, supported and observed the development of national digital health strategies and has monitored progress through surveys and global maturity models. WHO has advocated for the use of information and communication technology (ICT) in health through several resolutions since 2005, when it adopted resolution WHA58.28 (4) urging countries and stakeholders to create strategic plans for e-health and to establish e-health foundations (5). WHO has also proposed a number of tools and services to support strategic transformation towards digital health (6).

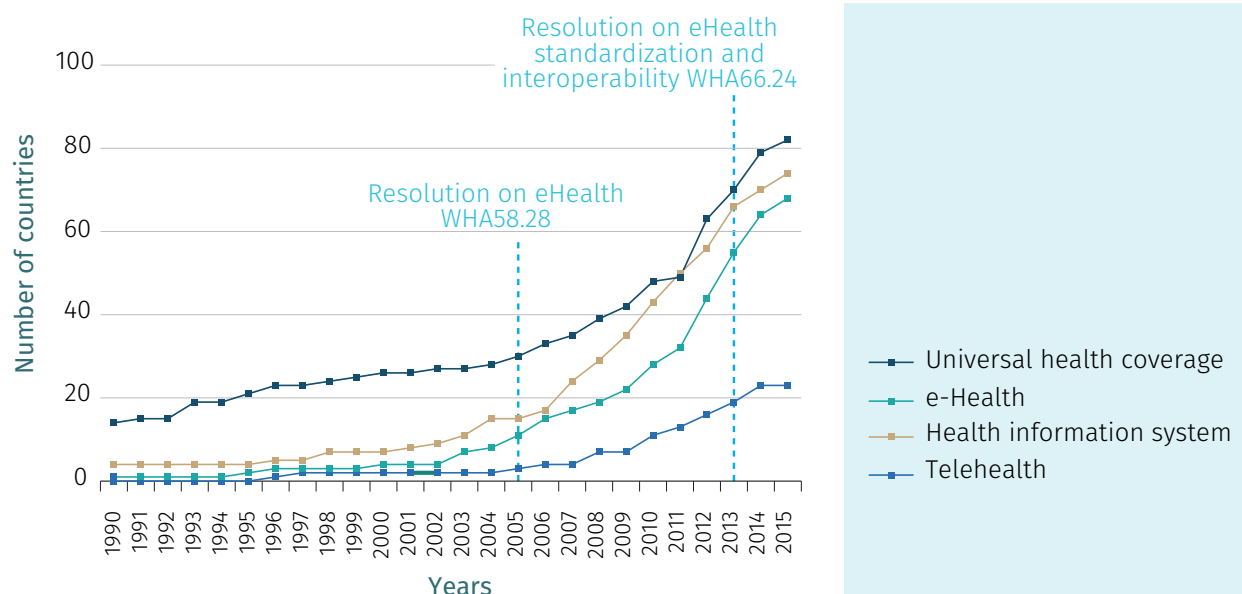
WHO's Global Observatory for e-Health (GOe) initiative was launched in 2005 to study the development and impact of e-health at a global level and to provide guidance on effective practices and standards in e-health. Through this initiative, global surveys on e-health were conducted between 2005 and 2013, providing countries with benchmarks to be used for the growth and development of their e-health services. As a result of this, country e-health profiles were published in 2011, 2013 and 2015 (7–9).

WHO, in collaboration with the International Telecommunication Union (ITU), introduced the National eHealth Strategy Toolkit in 2012 (10); this provides a three-stage framework for developing: 1) a national e-health vision; 2) an implementation roadmap reflecting country priorities and the e-health context; and 3) a progress monitoring plan to ensure impact and generate needed funding. WHO resolution WHA 66.24 (2013) emphasized the need for standardization and interoperability when adopting e-health solutions (11). It urged Member States to coordinate and collaborate to develop e-health roadmaps for implementation.

The Global Strategy on Digital Health 2020–2025 was endorsed by the Seventy-Third World Health Assembly in decision WHA73(28) (2020) (12). The strategy was developed as a response to resolution WHA71.7 of May 2018, which recognized the role of digital health in enhancing health service delivery models, strengthening people-centred health services and contributing to improved population health and health equity (13). The strategy underscores the potential support offered by digital technologies for patient-centred care and UHC, including health promotion, protection, treatment, diagnosis, rehabilitation and palliative care without financial burden on individuals, families and communities.

Fig. 1 illustrates the cumulative level of adoption of digital health over the years 1990–2015. It also shows the impact of WHO resolutions on upward trends in digital health, though many other factors also contributed to this penetration, including the affordability of technology and ease of use in application (14).

This report presents concise documentation of the digital health situation in the Region based on the regional survey conducted in 2021. Understanding the digital ecosystem in the Region will allow WHO to plan and support the needs of all its countries and pave the way for cost-effective adoption of appropriate technology according to country priorities and needs in the Region.



Source: Global diffusion of eHealth: making universal health coverage achievable: report of the third global survey on eHealth. Geneva: WHO; 2016 (<https://www.who.int/publications/i/item/9789241511780>). Licence: CC BY-NC-SA 3.0 IGO (14).

Fig. 1. Number of countries with UHC, e-Health, HIS, and tele-health policies or strategies, cumulatively by year of adoption, 1990–2015

Rationale for the regional survey

In 2021, WHO launched a plan to assist countries in the Region to adopt the *Global Strategy on Digital Health 2020–2025*. This raised a need to conduct an up-to-date situation analysis, while moving forward with strategic plans for digital health. The purpose of this regional survey was therefore to describe the current digital health landscape and implementation efforts to support Member States to strategically address national health priorities through well-established, people-centred digital health solutions and to monitor progress through standard measures compared with the baseline set by this study.

The survey was designed to enable each Member State to learn about opportunities and priorities to adopt the global strategy, with a vision of contributing to the achievement of UHC and the Sustainable Development Goals (SDGs). It was also intended to encourage a participatory approach and national stakeholder dialogues to help develop country visions, roadmaps and strategies for digital health.

Methodology

The survey was an updated version of the 2018 survey and included components from the WHO/ITU toolkit (10) for developing e-health strategies and the most recent comprehensive survey conducted by the GOe in 2015. The survey participants included WHO country offices and ministry of health focal points for e-health (or digital health).

The online web tool Datacol (in English) was used for data collection. This is a web-based survey platform owned by WHO; the same tool was used in the 2017–2018 survey. In 2021, a tool was developed and populated with baseline data (from the 2018 survey), after which the digital health focal points in each country were asked to validate and update the information with their feedback. The questionnaire in 2021–2022 had an additional section inquiring about the utilization of digital technologies as part of the COVID-19 response. The survey recorded data provided as answers to specific questions in a standardized way and included countries that had not responded previously. The purpose was to provide a broader perspective highlighting challenges at the regional level.

The tool was piloted in two Member States (Morocco and Tunisia) in collaboration with digital health focal points. Subsequently, based on the feedback, the tool was enhanced to clarify some sections and add descriptions.

The cross-sectional survey was conducted among the 22 countries/territories of the Eastern Mediterranean Region in 2021 and 2022. Respondents were designated digital health focal points in countries/territories, as well as national leaders nominated by ministries of health or researchers in e-health in the targeted countries. After implementation of the survey, the country focal points were invited to participate in capacity-building workshops on developing national e-health (or digital health) strategies and action plans.

The survey assessed most of the core components of digital health referred to in the WHO/ITU toolkit and further developed in the Global strategy on digital health to include: 1) leadership and governance; 2) investment and operations; 3) services and applications for scaling up; 4) integration and sustainability; 5) respect for standards and interoperability; 6) a flexible digital infrastructure; 7) an adaptable health workforce; 8) legislation, ethics policies and compliance; and 9) a people-centred approach.

The survey enabled comprehensive country profiles to be developed for each Member State (Annex 1) to present an overview of individual digital health progress in all countries/territories of the Region. Each profile contains a combination of data from WHO country profiles and from the survey results. Each country profile shows progress made in developing a national digital health strategy along with workforce development, standards and interoperability, as well as digital health applications currently present in the country. The survey responses were cross-checked against publicly available data for quality and assurance purposes (Annex 2).



Results

Survey responses

All 22 countries/territories in the Region submitted their feedback, with a response rate of 100% to the 2021–2022 survey on digital health.

Section 1. Foundations of digital health

Foundations of digital health are components that enable the establishment of sustainable digital health solutions. Four foundation components were addressed in this survey: 1) setting health priorities (investment determinants); 2) vision for digital transformation that reflects the commitment of the country (including five indicators for strategic commitment to use ICT in addressing health priorities); 3) legislation; and 4) capacity-building of health professionals on ICT for health.

1. Health priorities

A wide range of health priorities were identified by countries, ranging from management of disease surveillance to health information strengthening and broader areas of UHC. These are based on a country's maturity in digital health, its priorities and needs, and understanding of the use of digital health in health care. A full list of priorities submitted by countries is given in Annex 3.

2. Digital transformation commitment (policies and strategies)

The survey follows the same approach adopted by the most recent GOe survey from 2015, which considered the development of both strategies and policies as commitment to a strategic move towards digital transformation. In this section, five questions were asked regarding commitment to digital health strategies and policies. Seven countries (31%) reported having an established policy or strategy for e-health or digital health, while eight countries (36%) were in the process of developing a policy or strategy of this kind. Only five countries (22%) in the Region did not have a national health policy or strategy that clearly refers to the use of ICT or e-health. Table 1 synthesizes the responses from Member States. The leading countries identified with commitment in all five areas of e-health were Bahrain, Islamic Republic of Iran, Saudi Arabia and Sudan.

Of the 22 countries/territories, 14 countries (63.6%) reported the existence of a national health policy or strategy that clearly refers to the use of ICT or e-health, which was the highest level of commitment from Member States of the five categories. On the other hand, only seven countries (31.8%) reported having a national e-health (or digital health) policy or strategy, which was the lowest level of commitment from Member States of the five categories.

3. Legislation for digital applications

The development of legislation, awareness of digital applications and full implementation are core elements for the safe adoption of digital health systems. Without legal cover, the adoption of digital health solutions could be risky and might have adverse impacts on patients and health service providers. Table 2 shows that 15 (68.2%) countries in the Region have developed legislation to protect privacy and personally identifiable data of individuals. This is just a sample of a core set of laws and policies that establishes the foundation for mature adoption of digital health services. This shows an increase from 2016 survey where about half (57%) of the countries in the Region reported having policies or legislation to protect the privacy of personally identifiable data of patients, either in paper or digital format.¹

4. Capacity-building for health workers on the use of ICT for health

As shown in Table 3, 16 (72.7%) countries have recognized and invested in development of the existing workforce.

¹ WHO. Atlas of eHealth country profiles: the use of eHealth in support of universal health coverage: based on the findings of the third global survey on eHealth 2015. World Health Organization; 2016 (<https://www.who.int/publications/i/item/9789241565219>).

Table 1. Survey responses of countries/territories regarding existing policy or strategy for e-health or digital health

Country	Does your country have a national health policy or strategy that clearly refers to the use of ICT or e-health?	Does your country have a national universal health coverage policy or strategy that clearly refers to the use of ICT or e-health?	Does your country have a national e-health (or digital health) policy or strategy?	Does your country have a national health information system (HIS) policy or strategy?	Does your country have strategies to establish electronic health information exchange (ICT infrastructure and core services)?
Afghanistan	Yes	No	In process	Yes	Yes
Bahrain	Yes	Yes	Yes	Yes	Yes
Djibouti	No	No	No	In process	In process
Egypt	No	In process	Yes	Yes	In process
Iran (Islamic Republic of)	Yes	Yes	Yes	Yes	Yes
Iraq	Yes	In process	No	In process	In process
Jordan	Yes	Don't know	No	Yes	No
Kuwait	In process	In process	In process	In process	In process
Lebanon	Yes	Yes	In process	In process	In process
Libya	No	No	In process	No	No
Morocco	Yes	No	No	Yes	No
Oman	Yes	Yes	In process	Yes	Yes
Pakistan	Yes	Yes	In process	In process	No
Palestine	Yes	No	In process	Yes	No
Qatar	Yes	In process	Yes	Yes	Yes
Saudi Arabia	Yes	Yes	Yes	Yes	Yes
Somalia	In process	In process	In process	In process	In process
Sudan	Yes	Yes	Yes	Yes	Yes
Syrian Arab Republic	No	No	No	In process	In process
Tunisia	In process	Yes	No	No	In process
United Arab Emirates	Yes	Yes	Yes	In process	Yes
Yemen	No	No	No	Yes	No

Summary	Yes	14 (63.6%)	9 (40.9%)	7 (31.8%)	12 (54.5%)	8 (36.4%)
	In process	3 (13.6%)	5 (22.7%)	8 (36.4%)	8 (36.4%)	8 (36.4%)
	No	5 (22.7%)	7 (31.8%)	7 (31.8%)	2 (9.1%)	6 (27.3%)
	Don't know	0 (0%)	1 (4.5%)	0 (0%)	0 (0%)	0 (0%)

Table 2. Survey responses of countries/territories regarding legislation for digital health applications

Country	Does your country have legislation(s) to protect the privacy of personally identifiable data of individuals, either in paper or digital format?
Afghanistan	Yes
Bahrain	Yes
Djibouti	No
Egypt	Yes
Iran (Islamic Republic of)	Yes
Iraq	Yes
Jordan	Yes
Kuwait	Yes
Lebanon	In process
Libya	Yes
Morocco	Yes
Oman	In process
Pakistan	Yes
Palestine	Yes
Qatar	In process
Saudi Arabia	Yes
Somalia	No
Sudan	Yes
Syrian Arab Republic	Yes
Tunisia	Yes
United Arab Emirates	In process
Yemen	Don't know

Table 3. Survey responses of countries/territories regarding capacity-building for digital health

Country	Does your country provide in-service training for capacity-building of health professionals in the use of ICT for health?
Afghanistan	No
Bahrain	Yes
Djibouti	No
Egypt	Yes
Iran (Islamic Republic of)	Yes
Iraq	Yes
Jordan	Don't know
Kuwait	Yes
Lebanon	Yes
Libya	Yes
Morocco	No
Oman	Yes
Pakistan	Yes
Palestine	Yes
Qatar	Yes
Saudi Arabia	Yes
Somalia	No
Sudan	Yes
Syrian Arab Republic	Yes
Tunisia	Yes
United Arab Emirates	Yes
Yemen	No

Summary	Yes	16 (72.7%)
	In process	0
	No	5 (23%)
	Don't know	1 (4%)

Section 2. Digital health services and applications

This section maps the existence of basic digital health applications in countries of the Region. Five categories of digital health applications were considered, as outlined in Table 4.

The United Arab Emirates and Sudan reported the highest number of digital health applications adopted overall, followed by Bahrain.

Looking more closely, Islamic Republic of Iran, Sudan and United Arab Emirates reported the highest levels of adoption under the first category, “Individual electronic health information”. For both “Health care communication and collaboration” and “Health care service delivery tools”, Bahrain, Sudan and the United Arab Emirates reported the highest level of adoption for communication and collaboration purposes. Qatar, Sudan and the United Arab Emirates were the top countries reporting the adoption of digital health for “Health information and knowledge”. Similarly, the adoption of digital health applications for health care management and administration was highest in Bahrain, Sudan, Qatar and the United Arab Emirates.

Table 4. Survey responses of Member States regarding digital health services and applications

Digital health application scope	Application component	Afghanistan	Bahrain	Djibouti	Egypt	Iran (Islamic Republic of)	Iraq	Jordan	Kuwait	Lebanon	Libya	Morocco	Pakistan	Palestine	Qatar	Saudi Arabia	Somalia	Sudan	Syrian Arab Republic	Tunisia	United Arab Emirates	Oman	Yemen	Total	% of countries
Individual electronic health information	Electronic medical records	-	✓	-	✓	✓	-	✓	✓	-	-	-	-	✓	✓	✓	-	✓	-	✓	✓	✓	-	12	55.0%
	Personal health records	-	-	-	-	✓	-	-	-	-	-	-	-	-	-	-	-	✓	-	-	✓	-	-	3	14.0%
	Electronic health records	-	✓	-	-	✓	-	-	-	-	-	-	-	-	-	✓	-	✓	-	-	✓	✓	-	6	27.0%
Health care communication and collaboration	Access to electronic health records and test results	-	✓	-	-	✓	-	-	✓	-	-	-	-	-	✓	✓	-	✓	-	✓	✓	✓	-	9	41.0%
	Appointment booking and management	-	✓	-	-	-	-	✓	-	-	-	-	-	-	✓	✓	-	✓	-	-	✓	-	-	6	27.0%
	Care plan management	-	✓	-	-	-	-	-	-	-	-	-	-	-	-	-	-	✓	-	-	✓	-	-	3	14.0%
	Electronic health event summaries	-	✓	-	-	✓	-	-	-	-	-	-	-	-	✓	-	-	✓	-	✓	✓	✓	-	7	32.0%
	Electronic referrals and specialist letters	-	✓	-	-	✓	-	-	-	-	-	-	-	-	-	✓	-	✓	-	-	✓	✓	-	6	27.0%
	Health care provider/ service directories	-	✓	-	-	-	✓	✓	-	-	-	✓	-	-	-	✓	-	✓	-	-	✓	✓	-	8	36.0%

Table 4. Survey responses of Member States regarding digital health services and applications (*continued*)

Digital health application scope	Application component	Afghanistan	Bahrain	Djibouti	Egypt	Iran (Islamic Republic of)	Iraq	Jordan	Kuwait	Lebanon	Libya	Morocco	Pakistan	Palestine	Qatar	Saudi Arabia	Somalia	Sudan	Syrian Arab Republic	Tunisia	United Arab Emirates	Oman	Yemen	Total	% of countries
Health care service delivery tools	Alerts monitoring and management	-	✓	-	-	-	✓	✓	-	-	-	✓	-	-	✓	-	-	✓	-	-	✓	✓	-	8	36.0%
	Chronic disease management	-	✓	-	-	✓	-	-	-	-	-	-	-	-	-	✓	-	✓	-	-	✓	-	-	5	23.0%
	Clinical decision support	-	✓	-	-	-	-	-	-	-	-	-	-	-	✓	-	-	✓	-	-	✓	✓	-	5	23.0%
	Medications management	-	✓	-	-	-	-	-	✓	-	-	✓	-	-	✓	✓	-	✓	-	✓	✓	✓	-	9	41.0%
	Prescription decision support	-	✓	-	-	✓	-	-	-	-	-	-	-	-	✓	✓	-	✓	-	-	✓	✓	-	7	32.0%
	Real-time clinical data analysis	-	✓	-	-	-	-	✓	-	-	-	-	-	-	-	-	-	✓	-	-	✓	-	-	4	18.0%
	Telemedicine and m-health	✓	✓	-	✓	✓	-	✓	-	-	-	-	✓	✓	✓	✓	-	✓	-	✓	✓	-	-	12	55.0%
Health information and knowledge	Consumer health knowledge sources	-	-	-	✓	-	-	-	-	-	-	-	-	-	✓	-	-	✓	-	-	✓	-	-	4	18.0%
	Distance learning/ electronic resources	-	✓	-	-	✓	-	-	-	-	-	✓	-	✓	✓	✓	-	✓	-	-	✓	✓	-	9	41.0%
	Health care provider knowledge sources	-	-	-	-	✓	✓	-	-	-	-	-	-	-	✓	✓	-	✓	-	-	✓	-	-	6	27.0%
Health care management and administration	Adverse event monitoring	-	✓	-	✓	✓	-	-	-	-	-	✓	-	-	✓	-	-	✓	-	-	✓	✓	-	8	36.0%
	Clinical practice improvement	-	✓	-	-	-	-	-	-	-	-	-	-	-	✓	-	-	✓	-	-	✓	-	-	4	18.0%
	Compliance monitoring	-	✓	-	-	-	-	-	-	-	-	-	-	-	-	✓	-	✓	-	-	-	✓	-	4	18.0%
	Health care and clinical research	-	✓	-	-	-	-	-	-	-	-	-	-	-	✓	✓	-	-	-	-	✓	-	-	4	18.0%
	Health policy development	-	✓	-	-	-	✓	-	-	-	-	-	✓	-	✓	-	-	-	-	-	✓	-	-	5	23.0%
	Operations management	-	✓	-	-	-	-	-	-	-	-	-	-	-	-	✓	-	✓	-	✓	✓	-	-	5	23.0%
	Risk analysis	-	-	-	-	✓	-	-	-	-	-	-	-	-	-	-	-	✓	-	-	-	✓	-	3	14.0%
	Surveillance	-	✓	-	-	✓	-	✓	-	-	-	✓	-	✓	✓	✓	-	✓	-	-	-	✓	-	9	41.0%

Fig. 2 shows the penetration of digital health applications in countries/territories of the Region. Applications related to the “Individual electronic health information” category (such as electronic medical records) and applications related to “Health care service delivery tools” (such as telemedicine, m-health and medications management) were the most widely reported in the Region. For example, telemedicine and m-health (service delivery tools) and electronic medical records (electronic health information) have reportedly been adopted in 55% of countries in the Region. Meanwhile, medications management, distance learning and surveillance applications were reported to have been adopted in 41% of countries.

It is to be noted that 41% of countries claimed to have access to electronic health records and test results while these records are available in only 27% of countries of the Region. Some validation of the meaning of this feedback is therefore required.

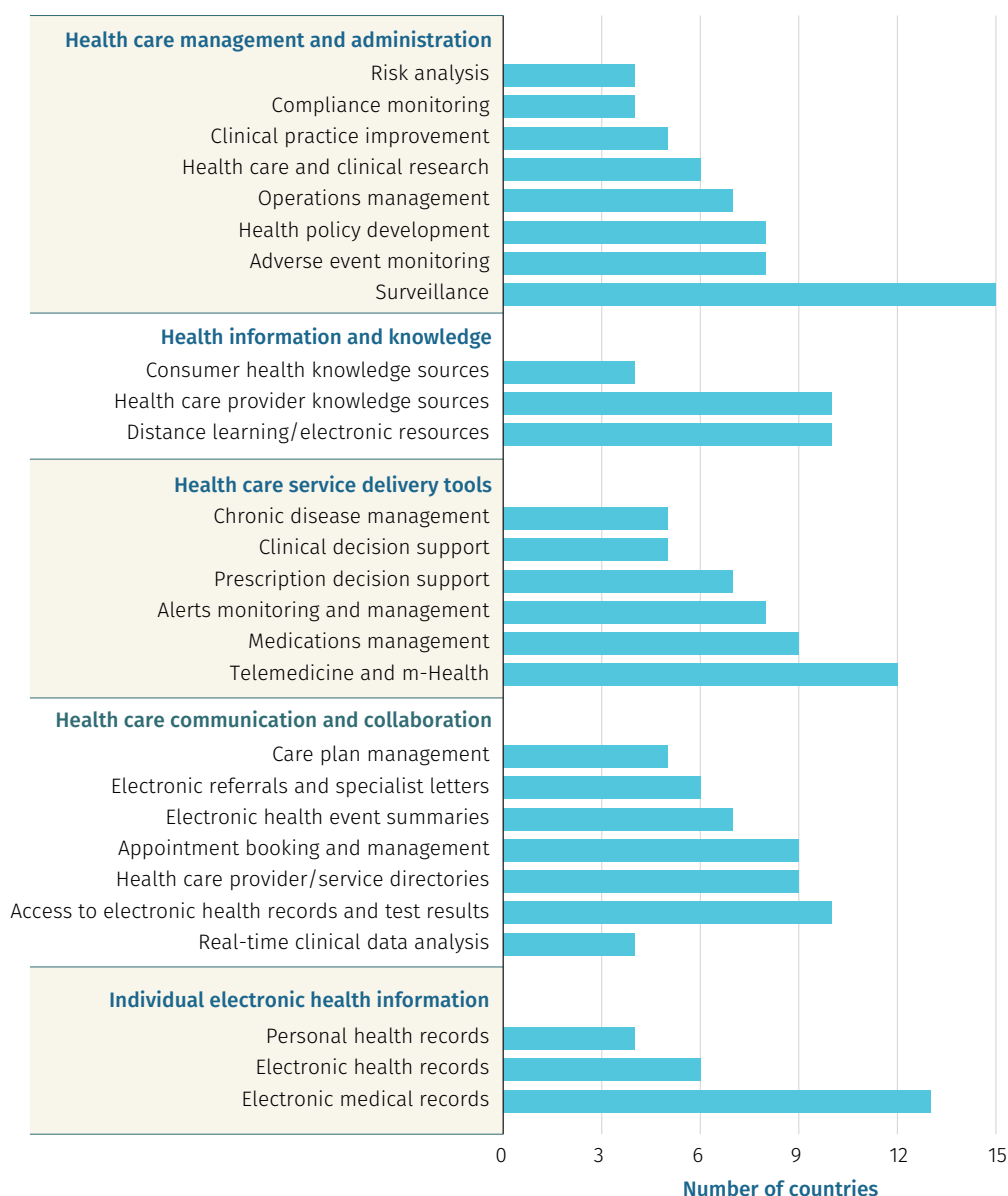


Fig. 2. Number of countries having digital health applications and services

Section 3. Digital health standards and interoperability

For digital health applications to be effective in supporting health systems and services, there is a need for a high level of interoperability and standardization. Fifteen countries (68.0%) claimed to have at least one component of standardization. The highest level of adoption of digital health standardization was reported to be in the category of “standard terminologies” (to enable information to be communicated electronically in a similar way for describing symptoms, diagnoses and treatments e.g. clinical coding standards), with 12 countries (55.0%) reporting the adoption of such standards. The second highest was in “data structure standards” (which govern how datasets are stored to enable aggregation and presentation of complete data), reported to be available in nine countries (41.0%). The third highest was in “software accreditation standards” (used to enforce interoperability of software used with other services in health systems), reported in seven countries (32.0%) (see Table 5).

Table 5. Survey responses of countries/territories regarding digital health standards and interoperability

Standardization components	Afghanistan	Bahrain	Djibouti	Egypt	Iran (Islamic Republic of)	Iraq	Jordan	Kuwait	Lebanon	Libya	Morocco	Pakistan	Palestine	Qatar	Saudi Arabia	Somalia	Sudan	Syrian Arab Republic	Tunisia	United Arab Emirates	Oman	Yemen	Total	Percentage of countries
Data structure standards	✓	✓	–	–	✓	–	–	–	✓	–	–	✓	–	✓	✓	–	–	–	–	✓	✓	–	9	41.0%
Common terminology standards	–	✓	–	–	✓	–	✓	–	✓	–	✓	–	✓	✓	✓	–	–	✓	✓	✓	✓	–	12	55.0%
Software accreditation standards	–	✓	–	–	✓	–	–	–	–	–	–	–	–	✓	✓	–	✓	✓	–	–	✓	–	7	32.0%
Summary	✓	✓	–	–	✓	–	✓	–	✓	–	✓	✓	✓	✓	✓	–	✓	✓	✓	✓	✓	–	15	68.0%

Section 4. National governance and guidance of digital health

Effective governance and visionary leadership are key in integrating stakeholders’ efforts to regulate and monitor the ethical and safe adoption of technology and to sustain continual improvements in health systems through digital health. The survey feedback on the question of governance was synthesized into five categories, with 14 countries/territories (64%) of the Region falling into the first two categories. In the first category, the country’s ministry of health was reported to govern and lead the implementation of digital health (eight countries, representing 38.0%; $n=21$ countries that responded to this question). The second category included countries where digital health is still led by the ministry of health but a further committing step has been taken by setting up a dedicated structure (a department or a committee) within the ministry to lead (six countries, representing 29.0% of responding countries). Four countries reported that they allocated multisectoral ministerial leaders to coordinate the governance of digital health (18.0% of reporting countries). Afghanistan and Yemen reported relying on WHO and nongovernmental organizations to guide and advocate for the implementation of digital health. Only Qatar reported that it relies on a public–private partnership model, through a hybrid national committee that assesses priorities and proposes strategies, policies and investment decisions for leveraging the use of digital health in supporting the health system (judging the effectiveness of this requires an impact assessment). Fig. 3 shows the number of countries in each of the five categories.

Table 6 groups countries under each of the governance categories; this could be used as a reference for facilitating dialogue with stakeholders within countries that plan to develop and implement strategic plans for digital health implementation.



Fig. 3. Survey responses of Member States regarding digital health governance

Table 6. Survey responses of Member States regarding digital health governance

Governance category	Digital health governing entity
Ministry of health non-specified structure	
Bahrain	Ministry of Health
Egypt	No advisory board is available
Iraq	Ministry of Health
Morocco	Ministry of Health
Oman	Ministry of Health
Palestine	Ministry of Health
Somalia	Ministry of Health Federal Government of Somalia
United Arab Emirates	Ministry of Health and Prevention
Ministry of health specified structure	
Djibouti	INSTAD Statistics Department of Djibouti
Lebanon	e-Health Unit at the Ministry of Public Health
Libya	Health information centre
Syrian Arab Republic	Ministry of Health IT department
Tunisia	Centre Informatique du Ministère de la Santé and e-Health Unit in the Ministry of Public Health
Multisectoral ministerial leaders	
Iran (Islamic Republic of)	Minister's Directorate for Statistics and ICT/Minister's Health Observatory Department/HIS Department of Public Health
Jordan	E-Transformation and Information Technology Director
Pakistan	Technical Working Group on e-Health
Sudan	Federal Ministry of Health/National Information Centre in the Ministry of Communication and Information Technology
Public-private partnership	
Qatar	National e-Health Committee
Saudi Arabia	Ministry of Health, Saudi Health Council and Saudi Data and Artificial Intelligence Authority (SDAIA)
WHO	
Afghanistan	WHO
Yemen	WHO
No information	
Kuwait	No response

Section 5. Perceived barriers to adopting and implementing e-health programmes

Despite the commitment of most Member States towards digital health, implementation faces many challenges and barriers. In this survey countries were asked to identify, in order of importance, the most significant barriers to implementing digital health solutions. In Table 7, the frequency with which a barrier was reported as one of the top three challenges was used to weight the order in which barriers are shown. The weight column is calculated as the number of countries reporting a barrier, multiplied by the importance that the country gave to that particular barrier.

The barrier to the implementation of digital health reported most frequently by countries was “capacity of workforce”, which refers to the scarcity of qualified health professionals, as well as leaders who could manage the adoption of digital health solutions. The second biggest barrier was “funding”, which refers to allocating the investments needed to enable digital health implementation. The third biggest barrier was “competing priorities”. The fourth biggest barrier, “lack of infrastructure”, was also recognized by most countries, and could be considered a result of the first three barriers.

Section 6. Digital response to the COVID-19 pandemic

Barriers relevant to digital solutions during the pandemic

Thirteen countries (62.0%) reported difficulties when planning to implement digital solutions during the pandemic. It is important to compare challenges before the pandemic and during the implementation of COVID-19 applications, noting that the serious threats to life posed by the pandemic exposed many failures in health systems. Table 8 synthesizes the feedback reported on challenges faced by countries when adopting new technologies.

While the first challenge (funding) is similar to barriers reported prior to the pandemic, the urgent need to introduce technology to save lives exposed an existing dimension, which was lack of “expertise”. It takes several years of cumulative practical use of a technology to be able to identify hidden dimensions of complexity and resistance to change. Capacity-building to produce trained staff with sufficient knowledge was not adequate to implement new technology or to sustain or govern its impact. A substantial degree of institutionalization and practical governance are required to move forward with such strategic development. Meanwhile, new challenges emerged during the pandemic, including the selection and adoption of reliable and easy-to-use software with appropriate accreditation; managing the culture to accept new modalities for health services; strategic planning; managing the fragmentation of duplicate investments; and dealing with language barriers for cross-border applications and for expatriate service users.

Table 7. Survey responses of Member States regarding barriers to the implementation of digital health

Implementation barrier	Frequency of reported importance (5=high importance)					Weight
	5	4	3	2	1	
Capacity: lack of trained human resources and/or technical support	6	4	7	2	2	73
Funding: lack of funding to develop and support services	3	8	7	2	0	72
Priorities: competing health system priorities	7	2	5	2	2	64
Infrastructure: lack of equipment and/or connectivity	2	5	9	2	1	62
Legal: lack of legislation or regulations covering digital health programmes	5	5	3	3	0	60
Cost-effectiveness: lack of evidence on cost-effectiveness of programmes	5	4	2	4	1	56
Policy: national policies do not recognize digital health in health services delivery	5	2	4	4	2	55
Demand: lack of demand for digital health programmes by health professionals or target groups	3	1	5	1	8	44
Effectiveness: lack of evidence on effectiveness of programmes	2	5	1	4	2	43

Digital health applications relevant to the pandemic

Table 9 summarizes the reported digital health applications that countries in the Region adopted in response to the COVID-19 pandemic. The most frequently reported application, as reported by 15 countries (68%), was tools for monitoring the vaccination administration management system. The second most commonly reported application was telemedicine/telehealth, which supported remote assessment of patients and helped to reduce avoidable face-to-face contact between patients and health care providers. Eight countries (36%) reported the use of telehealth during the pandemic. Diagnostic applications also benefited from artificial intelligence and other technologies to identify distinguishing signs of the disease. Digital contact tracing also represented a paradigm shift in many countries and helped to introduce retrospective tracking that was not biased by positive cases when patients were asked to remember all persons with whom they had been in contact.

Annex 1 presents a summary of the digital health transformation process in each country/territory. The country profiles show that all countries/territories are embracing digital health in various capacities, and also provide side-by-side comparison of the areas of excellence and areas where further development is needed in each country/territory.

Table 8. Survey responses of Member States regarding barriers to implementation of digital health solutions during COVID-19

Identified challenges	Frequency
Funding: insufficient budget allocation and infrastructure investments	8
Expertise: lack of technical capacity and trained workforce	7
Software: poor quality (not fit for purpose) and inadequate evaluation of performance	3
Culture: non-acceptance of digital health solution by populations/health workers	3
Planning: lack of strategic planning	1
Fragmentation: lack of coordination between government entities	1
Language: barriers especially with expatriate and displaced populations	1

Table 9. Survey responses of Member States regarding digital health applications used during the pandemic

Application	Number of countries
Vaccine administration management system	15
Telemedicine/telehealth	8
Diagnostic applications	6
Digital contact tracing	6
Chatbots	4
m-health	3
Hospital bed dashboard	1



Discussion

Foundations of digital health

Under the foundations of digital health, four key areas were discussed: health priorities; commitment to digital transformation; legislation for digital application; and capacity-building for health workers on the use of ICT for health.

All countries reported a maximum of five health priorities, which included control of communicable and noncommunicable diseases, reducing mortality, improving quality of health services, strengthening routine health information systems, strengthening primary health care and emergency response.

Most countries in the Region showed a commitment to digitalization, and this was reflected in health policies and/or strategies. In all, 63% of Member States have developed a national health policy or strategy that clearly refers to the use of ICT or e-health. Most barriers to the implementation of digital health plans were attributed to the lack of qualified capacity and resources.

For digital health to be successful, health data must be securely stored, retrieved and shared across authenticated stakeholders. Two thirds of Member States have already developed legislation to protect privacy and personally identifiable data of individuals, which is a foundational step towards developing digital health strategies.

Health care workers represent a core component of successful digital health implementation and utilization. However, fewer than half of the countries reported workforce development strategies for digital health. Only 40% of countries reported investment in continuous development of the existing health care workforce. It is imperative for countries to have digitally competent workforces that can adjust to new and emerging technologies, as well as to encourage and educate their patients to engage in digital health solutions.

Building a digital health workforce requires continuous training and education for workers to remain engaged and informed regarding the latest best practices in digital health. To support the commitment of countries to digital health, there is a critical need to improve the skills and abilities of health care workers on a regular basis. Examples of workforce development include educational webinars, training workshops and accredited continuing education programmes. The implementation of continuous professional development and training programmes will enable health care workforces to cope with emerging digital health solutions.

Digital health services and applications

More than half of Member States reported the adoption and utilization of telemedicine, m-health and electronic medical records. The use of telemedicine and m-health can bridge current health disparities, improve outcomes and reduce health care-related costs. This was clearly demonstrated during the response to the COVID-19 pandemic. Meanwhile, electronic medical records are crucial for the seamless storage and retrieval of health data to improve health care delivery, as well as the rational use of resources and health service utilization.

The use of personal health records was low, being reported by only three Member States. Personal health records can help patients to better manage their care, even when public systems are not yet effectively developed. Providing access to important health information – such as laboratory results, immunization records and screening due dates – in electronic format updating and sharing patients' records across health service providers is made easier. Therefore, personal health records allow patients to be more engaged with their health providers, enhance patient/provider communication and thus reduce administrative burdens.

Digital health standards and interoperability

More than half of countries reported the adoption of standard terminologies to enable information to be communicated electronically for describing symptoms, diagnoses and treatments. Only a third of Member States reported using software accreditation standards to ensure that the digital health solutions adopted would achieve their goals. Accreditation standards are important in validating the design and functionality of digital health solutions to enforce best practices and to yield high levels of adoption by end-users. Accredited digital health solutions typically examine criteria around privacy, security, usability, interoperability and error recovery. Therefore, it is paramount for Member States to meet digital health and software accreditation standards to ensure that the software they choose provides a high-quality and reliable service.

About 41% of Member States reported recognition and thus utilization of data structure standards. Such standards govern the way that health datasets are stored and are used to ensure that information is interpreted consistently when exchanged and is efficiently aggregated for sound decision-making.

Perceived barriers to adopting and implementing digital health programmes

Two main barriers to the adoption and implementation of digital health applications were reported: 1) a lack of trained cadres in supporting functions; and 2) insufficient funding. The lack of trained human resources and technical support for digital health constitutes a major challenge to Member States aspiring to expand their utilization of digital health technology. Digital health capacity-building can be further improved through training workshops provided by technical and clinical experts; digital literacy programmes that advance workforce skills and knowledge; and advocacy that ensures the continuity of workforce development programmes.

Allocating sufficient funding to develop and sustain digital health programmes remains a major challenge for many Member States. Inadequate funding hinders the ability to build robust digital health infrastructure, train workforces, conduct public health campaigns with regards to digital health, and implement and adopt digital health solutions. Identifying sustainable funding sources and capital investments to support the development of digital health initiatives and systems is thus needed, and may be achieved through robust public-private partnerships.



Recommendations

Member States need to demonstrate increased commitment by developing national e-health strategies and health information policies in alignment with regional and global digital health strategies.

National strategies should address the core components of the digital health ecosystem, including electronic health records, telemedicine, standardization, interoperability, infrastructure, capacity-building and funding.

The third objective under the Global Digital Health Strategy (and the third objective of the Eastern Mediterranean regional strategy) is to build people-centred rather than hospital-centred health systems. This requires a commitment by all Member States to UHC, oriented towards strengthening primary health care.

National strategies need to be created for development of the health workforce that outline ways to engage health workers in education and training to ensure continuous professional development and the acquisition of digital health skills.

Countries need to outline regulations that enable the advancement of interoperability, prohibit information blocking and enhance the usability, accessibility, privacy and security of digital health services.

Member States need to assess the digital technology needs of stakeholders (health workers, patients, organizations and government), which includes the solicitation of end-user perceptions of the usability and acceptance of technologies and ethical considerations, as well as prevalent value systems.

Multilevel implementation strategies need to be developed which tailor systems to support safe, high-quality patient care and ensure that contingency plans are established to address system downtimes.



Conclusion

This report demonstrates that, post-COVID-19, all countries/territories in the Region have accelerated their efforts in transforming digital health. Although some countries have developed national strategies, most are facing impediments to implementation, such as socioeconomic disparities, conflicts and population displacements on top of technological, financial, legal, ethical and cultural challenges. In many countries strategic direction is still lacking to fully realize the potential of digital health. Countries with relatively advanced foundations, legislation and policies for the use of digital health services were able to implement solutions in support of the COVID-19 response more effectively. These solutions included strategies that aimed to enhance access to health care at times when extensive social distancing policies were being implemented. Such countries were able to optimize the use of limited resources to serve the people in greatest need and hence save lives. Others need to further enhance efforts to develop digital health strategies, action plans with roadmaps, national digital health architecture and investment plans. They should allocate resources and encourage national investment in standards-based and interoperable digital health systems at both national and district levels. There is a need for national capacity-building for digital health and related innovations, linked to country health and public health priorities.

WHO's regional strategy for fostering digital health in the Eastern Mediterranean Region (2023–2027) helps to provide the vision and goals for digital health and can be adapted by all countries to develop roadmaps and action plans in digital health according to their own resources, priorities and values, towards improving the health care and well-being of their populations. The country profiles in Annex 1 can be used to identify areas which need further attention and resource allocation in health and development plans relating to digital transformation and supporting digital health in different Member States of the Region.

WHO will provide technical support and relevant resources for the development of national action plans and the setting of norms and standards, and will coordinate the involvement of relevant United Nations agencies and stakeholders in priority digital health programmes.



References¹

1. Digital health: A strategy to maintain health care for people living with noncommunicable diseases during COVID-19. PAHO; 2022 (<https://iris.paho.org/handle/10665.2/52543>).
2. Mandil A, Mabry R, Milani B, Nour M, Afifi M, Abdel-Ghani K. Mapping of health innovations in response to the COVID-19 pandemic in Eastern Mediterranean and selected Arab Countries. *East Mediterr Health J.* 2022;28(2):130–143. <https://doi.org/10.26719/emhj.22.028>.
3. Global strategy on digital health 2020–2025. Geneva: World Health Organization 2021 (<https://www.who.int/docs/default-source/documents/gsdhdaa2a9f352b0445bafbc79ca799dce4d.pdf>). Licence: CC BY-NC-SA 3.0 IGO.
4. Fifty-eighth World Health Assembly, Geneva, 25 May: resolution WHA58.28: eHealth. Geneva: World Health Organization 2005 (https://apps.who.int/gb/ebwha/pdf_files/WHA58/WHA58_28-en.pdf).
5. Building foundations for eHealth: progress of member states: report of the Global Observatory for eHealth. WHO Global Observatory for e-Health; 2006 (<https://apps.who.int/iris/handle/10665/43599>).
6. Directory of eHealth policies survey. In: World Health Organization/Global Observatory of e-Health [website]. Geneva: WHO; 2024 (<https://www.who.int/observatories/global-observatory-for-ehealth/policies>).
7. Atlas eHealth country profiles: based on the findings of the second global survey on eHealth. Geneva: WHO; 2011 (https://www.afro.who.int/sites/default/files/2017-06/AHO-goe_atlas_2010_0.pdf).
8. Atlas of eHealth country profiles: eHealth and innovation in women's and children's health. Geneva: WHO; 2013 (<https://www.afro.who.int/publications/atlas-ehealth-country-profiles-ehealth-and-innovation-womens-and-childrens-health>).
9. Atlas of eHealth country profiles: the use of eHealth in support of universal health coverage: report of the third global survey on eHealth. Geneva: WHO; 2016 (<https://www.who.int/publications/i/item/9789241565219>).
10. World Health Organization and International Telecommunication Union. National eHealth Strategy Toolkit. WHO/ITU; 2012 (https://iris.who.int/bitstream/handle/10665/75211/9789241548465_eng.pdf?sequence=1).
11. Sixty-sixth World Health Assembly Geneva, 27 May 2013: resolution WHA66.24: eHealth standardization and interoperability Geneva: World Health Organization; 2013 (<https://apps.who.int/iris/handle/10665/150175>).
12. Seventy-third World Health Assembly, 13 November 2020: decision WHA73(28): Global strategy on digital health. Geneva: World Health Organization; 2020 ([https://apps.who.int/gb/ebwha/pdf_files/WHA73/A73\(28\)-en.pdf](https://apps.who.int/gb/ebwha/pdf_files/WHA73/A73(28)-en.pdf)).
13. Seventy-first World Health Assembly 26 May 2018: decision WHA71.7: Digital health. Geneva: World Health Organization; 2018 (https://apps.who.int/gb/ebwha/pdf_files/WHA71/A71_R7-en.pdf).
14. Global diffusion of eHealth: making universal health coverage achievable: report of the third global survey on eHealth. Geneva: World Health Organization; 2016 (<https://www.who.int/publications/i/item/9789241511780>).

¹ All references were accessed on 7 July 2024.



Annex 1. **Digital health country profiles**

Afghanistan

Country context			
World Bank: country population (millions)	41.129	World Bank: area (km ²)	652 230
World Bank: income group	Low income	ITU: individuals using the Internet (2020)	18.40%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	Yes	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	No	9 (40%)
National e-health (or digital health) policy or strategy	In process	7 (32%)
National health information system (HIS) policy or strategy	Yes	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	Yes	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	No	9 (41%)
Recognizing standardization and interoperability		
Data structure standards	Yes	9 (41%)
Common terminology standards	No	12 (55%)
Software accreditation standards	No	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	No	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	No	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	No	8 (36%)
Health care service delivery tools	Alerts monitoring and management	No	8 (36%)
	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	No	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	Yes	12 (55%)
Health information and knowledge	Consumer health knowledge sources	No	4 (18%)
	Distance learning/electronic resources	No	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
Health care management and administration	Adverse event monitoring	No	8 (36%)
	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	No	9 (41%)

Bahrain

Country context			
World Bank: country population (millions)	1.472	World Bank: area (km ²)	65
World Bank: income group	High income	ITU: individuals using the Internet (2020)	99.67%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	Yes	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	Yes	9 (40%)
National e-health (or digital health) policy or strategy	Yes	7 (31%)
National health information system (HIS) policy or strategy	Yes	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	Yes	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	Yes	9 (41%)
Common terminology standards	Yes	12 (55%)
Software accreditation standards	Yes	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	Yes	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	Yes	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	Yes	9 (41%)
	Appointment booking and management	Yes	6 (27%)
	Care plan management	Yes	3 (14%)
	Electronic health event summaries	Yes	7 (32%)
	Electronic referrals and specialist letters	Yes	6 (27%)
	Health care provider/service directories	Yes	8 (36%)
	Alerts monitoring and management	Yes	8 (36%)
Health care service delivery tools	Chronic disease management	Yes	5 (23%)
	Clinical decision support	Yes	5 (23%)
	Medications management	Yes	9 (41%)
	Prescription decision support	Yes	7 (32%)
	Real-time clinical data analysis	Yes	4 (18%)
	Telemedicine and m-health	Yes	12 (55%)
	Consumer health knowledge sources	No	4 (18%)
Health information and knowledge	Distance learning/electronic resources	Yes	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
	Adverse event monitoring	Yes	8 (36%)
Health care management and administration	Clinical practice improvement	Yes	4 (18%)
	Compliance monitoring	Yes	4 (18%)
	Health care and clinical research	Yes	4 (18%)
	Health policy development	Yes	5 (23%)
	Operations management	Yes	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	Yes	9 (41%)

Djibouti

Country context			
World Bank: country population (millions)	1.121	World Bank: area (km ²)	23 200
World Bank: income group	Lower middle income	ITU: individuals using the Internet (2020)	59.00%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	No	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	No	9 (40%)
National e-health (or digital health) policy or strategy	No	7 (31%)
National health information system (HIS) policy or strategy	In process	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	In process	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	No	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	No	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	No	9 (41%)
Common terminology standards	No	12 (55%)
Software accreditation standards	No	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	No	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	No	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	No	8 (36%)
Health care service delivery tools	Alerts monitoring and management	No	8 (36%)
	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	No	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	No	12 (55%)
Health information and knowledge	Consumer health knowledge sources	No	4 (18%)
	Distance learning/electronic resources	No	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
Health care management and administration	Adverse event monitoring	No	8 (36%)
	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	No	9 (41%)

Egypt

Country context			
World Bank: country population (millions)	110.990	World Bank: area (km ²)	1 002 450
World Bank: income group	Lower middle income	ITU: individuals using the Internet (2020)	71.91%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	No	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	In process	9 (40%)
National e-health (or digital health) policy or strategy	Yes	7 (31%)
National health information system (HIS) policy or strategy	Yes	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	In process	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	No	9 (41%)
Common terminology standards	No	12 (55%)
Software accreditation standards	No	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	Yes	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	No	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	No	8 (36%)
	Alerts monitoring and management	No	8 (36%)
Health care service delivery tools	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	No	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	Yes	12 (55%)
	Consumer health knowledge sources	Yes	4 (18%)
Health information and knowledge	Distance learning/electronic resources	No	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
	Adverse event monitoring	Yes	8 (36%)
Health care management and administration	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	No	9 (41%)

Iran (Islamic Republic of)

Country context			
World Bank: country population (millions)	88.551	World Bank: area (km ²)	1 648 195
World Bank: income group	Lower middle income	ITU: individuals using the Internet (2020)	75.57%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	Yes	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	Yes	9 (40%)
National e-health (or digital health) policy or strategy	Yes	7 (31%)
National health information system (HIS) policy or strategy	Yes	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	Yes	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	Yes	9 (41%)
Common terminology standards	Yes	12 (55%)
Software accreditation standards	Yes	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	Yes	12 (55%)
	Personal health records	Yes	3 (14%)
	Electronic health records	Yes	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	Yes	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	Yes	7 (32%)
	Electronic referrals and specialist letters	Yes	6 (27%)
	Health care provider/service directories	No	8 (36%)
Health care service delivery tools	Alerts monitoring and management	No	8 (36%)
	Chronic disease management	Yes	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	No	9 (41%)
	Prescription decision support	Yes	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	Yes	12 (55%)
Health information and knowledge	Consumer health knowledge sources	No	4 (18%)
	Distance learning/electronic resources	Yes	9 (41%)
	Health care provider knowledge sources	Yes	8 (36%)
Health care management and administration	Adverse event monitoring	Yes	8 (36%)
	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	Yes	3 (14%)
	Surveillance	Yes	9 (41%)

Iraq

Country context			
World Bank: country population (millions)	44.496	World Bank: area (km ²)	438 317
World Bank: income group	Upper middle income	ITU: individuals using the Internet (2019)	44.33%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	Yes	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	In process	9 (40%)
National e-health (or digital health) policy or strategy	No	7 (31%)
National health information system (HIS) policy or strategy	In process	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	In process	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	No	9 (41)%
Common terminology standards	No	12 (55)%
Software accreditation standards	No	7 (32)%

Existing digital health applications			
Individual electronic health information	Electronic medical records	No	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	No	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	Yes	8 (36%)
	Alerts monitoring and management	Yes	8 (36%)
Health care service delivery tools	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	No	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	No	12 (55%)
	Consumer health knowledge sources	No	4 (18%)
Health information and knowledge	Distance learning/electronic resources	No	9 (41%)
	Health care provider knowledge sources	Yes	8 (36%)
	Adverse event monitoring	No	8 (36%)
Health care management and administration	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	Yes	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	No	9 (41%)

Jordan

Country context			
World Bank: country population (millions)	11.286	World Bank: area (km ²)	89 342
World Bank: income group	Upper middle income	ITU: individuals using the Internet (2018)	65.2%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	Yes	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	Do not know	9 (40%)
National e-health (or digital health) policy or strategy	No	7 (31%)
National health information system (HIS) policy or strategy	Yes	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	No	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Don't know	9 (40%)
Recognizing standardization and Interoperability		
Data structure standards	No	9 (41%)
Common terminology standards	Yes	12 (55%)
Software accreditation standards	No	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	Yes	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	No	9 (41%)
	Appointment booking and management	Yes	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	Yes	8 (36%)
	Alerts monitoring and management	Yes	8 (36%)
Health care service delivery tools	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	No	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	Yes	4 (18%)
	Telemedicine and m-health	Yes	12 (55%)
	Consumer health knowledge sources	No	4 (18%)
Health information and knowledge	Distance learning/electronic resources	No	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
	Adverse event monitoring	No	8 (36%)
Health care management and administration	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	Yes	9 (41%)

Kuwait

Country context			
World Bank: country population (millions)	4.269	World Bank: area (km ²)	17 818
World Bank: income group	High income	ITU: individuals using the Internet (2020)	99.11%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	In process	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	In process	9 (40%)
National e-health (or digital health) policy or strategy	In process	7 (31%)
National health information system (HIS) policy or strategy	In process	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	In process	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	No	9 (41%)
Common terminology standards	No	12 (55%)
Software accreditation standards	No	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	Yes	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	Yes	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	No	8 (36%)
Health care service delivery tools	Alerts monitoring and management	No	8 (36%)
	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	Yes	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	No	12 (55%)
	Consumer health knowledge sources	No	4 (18%)
Health information and knowledge	Distance learning/electronic resources	No	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
	Adverse event monitoring	No	8 (36%)
Health care management and administration	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	No	9 (41%)

Lebanon

Country context			
World Bank: country population (millions)	5.490	World Bank: area (km ²)	10,452
World Bank: income group	Upper middle income	ITU: individuals using the Internet (2020)	84.10%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	Yes	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	Yes	9 (40%)
National e-health (or digital health) policy or strategy	In process	7 (31%)
National health information system (HIS) policy or strategy	In process	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	In process	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	In process	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	Yes	9 (41%)
Common terminology standards	Yes	12 (55%)
Software accreditation standards	No	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	No	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	No	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	No	8 (36%)
	Alerts monitoring and management	No	8 (36%)
Health care service delivery tools	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	No	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	No	12 (55%)
	Consumer health knowledge sources	No	4 (18%)
Health information and knowledge	Distance learning/electronic resources	No	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
	Adverse event monitoring	No	8 (36%)
Health care management and administration	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	No	9 (41%)

Country context			
World Bank: country population (millions)	6.812	World Bank: area (km ²)	1 759 540
World Bank: income group	Upper middle income	ITU: individuals using the Internet (2014)	17.76%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	No	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	No	9 (40%)
National e-health (or digital health) policy or strategy	In process	7 (31%)
National health information system (HIS) policy or strategy	No	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	No	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	No	9 (41%)
Common terminology standards	No	12 (55%)
Software accreditation standards	No	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	No	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	No	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	No	8 (36%)
Health care service delivery tools	Alerts monitoring and management	No	8 (36%)
	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	No	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	No	12 (55%)
Health information and knowledge	Consumer health knowledge sources	No	4 (18%)
	Distance learning/electronic resources	No	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
Health care management and administration	Adverse event monitoring	No	8 (36%)
	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	No	9 (41%)

Morocco

Country context			
World Bank: country population (millions)	37,458	World Bank: area (km²)	446 550
World Bank: income group	Lower middle income	ITU: individuals using the Internet (2020)	84.12%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	Yes	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	No	9 (40%)
National e-health (or digital health) policy or strategy	No	7 (31%)
National health information system (HIS) policy or strategy	Yes	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	No	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	No	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	No	9 (41%)
Common terminology standards	Yes	12 (55%)
Software accreditation standards	No	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	No	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	No	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	Yes	8 (36%)
	Alerts monitoring and management	Yes	8 (36%)
Health care service delivery tools	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	Yes	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	No	12 (55%)
	Consumer health knowledge sources	No	4 (18%)
Health information and knowledge	Distance learning/electronic resources	Yes	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
	Adverse event monitoring	Yes	8 (36%)
Health care management and administration	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	Yes	9 (41%)

Oman

Country context			
World Bank: country population (millions)	4.576	World Bank: area (km²)	309 500
World Bank: income group	High income	ITU: individuals using the Internet (2020)	95.23%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	Yes	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	Yes	9 (40%)
National e-health (or digital health) policy or strategy	In process	7 (31%)
National health information system (HIS) policy or strategy	Yes	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	Yes	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	In process	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	Yes	9 (41%)
Common terminology standards	Yes	12 (55%)
Software accreditation standards	Yes	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	No	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	No	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	No	8 (36%)
Health care service delivery tools	Alerts monitoring and management	No	8 (36%)
	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	No	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	Yes	12 (55%)
Health information and knowledge	Consumer health knowledge sources	No	4 (18%)
	Distance learning/electronic resources	No	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
Health care management and administration	Adverse event monitoring	No	8 (36%)
	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	Yes	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	No	9 (41%)

Pakistan

Country context			
World Bank: country population (millions)	235.825	World Bank: area (km ²)	881 912
World Bank: income group	Lower middle income	ITU: individuals using the Internet (2020)	25.00%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	Yes	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	Yes	9 (40%)
National e-health (or digital health) policy or strategy	In process	7 (31%)
National health information system (HIS) policy or strategy	In process	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	No	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	Yes	9 (41%)
Common terminology standards	No	12 (55%)
Software accreditation standards	No	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	Yes	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	No	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	No	8 (36%)
	Alerts monitoring and management	No	8 (36%)
Health care service delivery tools	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	No	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	Yes	12 (55%)
	Consumer health knowledge sources	No	4 (18%)
Health information and knowledge	Distance learning/electronic resources	Yes	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
	Adverse event monitoring	No	8 (36%)
Health care management and administration	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	Yes	9 (41%)

Palestine

Country context			
World Bank: country population (millions)	5.250	World Bank: area (km ²)	6220
World Bank: income group	Lower middle income	ITU: individuals using the Internet (2020)	74.60%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	Yes	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	No	9 (40%)
National e-health (or digital health) policy or strategy	In process	7 (31%)
National health information system (HIS) policy or strategy	Yes	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	No	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	No	9 (41%)
Common terminology standards	Yes	12 (55%)
Software accreditation standards	No	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	Yes	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	Yes	9 (41%)
	Appointment booking and management	Yes	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	Yes	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	No	8 (36%)
	Alerts monitoring and management	Yes	8 (36%)
Health care service delivery tools	Chronic disease management	No	5 (23%)
	Clinical decision support	Yes	5 (23%)
	Medications management	Yes	9 (41%)
	Prescription decision support	Yes	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	Yes	12 (55%)
	Consumer health knowledge sources	Yes	4 (18%)
Health information and knowledge	Distance learning/electronic resources	Yes	9 (41%)
	Health care provider knowledge sources	Yes	8 (36%)
	Adverse event monitoring	Yes	8 (36%)
Health care management and administration	Clinical practice improvement	Yes	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	Yes	4 (18%)
	Health policy development	Yes	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	Yes	9 (41%)

Qatar

Country context			
World Bank: country population (millions)	2.695	World Bank: area (km ²)	11 586
World Bank: income group	High income	ITU: individuals using the Internet (2020)	99.65%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	Yes	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	In process	9 (40%)
National e-health (or digital health) policy or strategy	Yes	7 (31%)
National health information system (HIS) policy or strategy	Yes	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	Yes	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	In process	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	Yes	9 (41%)
Common terminology standards	Yes	12 (55%)
Software accreditation standards	Yes	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	Yes	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	Yes	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	Yes	9 (41%)
	Appointment booking and management	Yes	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	Yes	6 (27%)
	Health care provider/service directories	Yes	8 (36%)
Health care service delivery tools	Alerts monitoring and management	No	8 (36%)
	Chronic disease management	Yes	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	Yes	9 (41%)
	Prescription decision support	Yes	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	Yes	12 (55%)
Health information and knowledge	Consumer health knowledge sources	No	4 (18%)
	Distance learning/electronic resources	Yes	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
Health care management and administration	Adverse event monitoring	No	8 (36%)
	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	Yes	4 (18%)
	Health care and clinical research	Yes	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	Yes	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	Yes	9 (41%)

Saudi Arabia

Country context			
World Bank: country population (millions)	36.409	World Bank: area (km ²)	2 149 690
World Bank: income group	High income	ITU: individuals using the Internet (2020)	97.86%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	Yes	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	Yes	9 (40%)
National e-health (or digital health) policy or strategy	Yes	7 (31%)
National health information system (HIS) policy or strategy	Yes	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	Yes	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	Yes	9 (41%)
Common terminology standards	Yes	12 (55%)
Software accreditation standards	Yes	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	No	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	No	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	No	8 (36%)
	Alerts monitoring and management	No	8 (36%)
Health care service delivery tools	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	No	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	No	12 (55%)
	Consumer health knowledge sources	No	4 (18%)
Health information and knowledge	Distance learning/electronic resources	No	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
	Adverse event monitoring	No	8 (36%)
Health care management and administration	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	No	9 (41%)

Somalia

Country context			
World Bank: country population (millions)	17,598	World Bank: area (km²)	637 657
World Bank: income group	Low income	ITU: individuals using the Internet (2017)	2.0%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	In process	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	In process	9 (40%)
National e-health (or digital health) policy or strategy	In process	7 (31%)
National health information system (HIS) policy or strategy	In process	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	In process	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	No	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	No	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	No	9 (41%)
Common terminology standards	No	12 (55%)
Software accreditation standards	No	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	Yes	12 (55%)
	Personal health records	Yes	3 (14%)
	Electronic health records	Yes	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	Yes	9 (41%)
	Appointment booking and management	Yes	6 (27%)
	Care plan management	Yes	3 (14%)
	Electronic health event summaries	Yes	7 (32%)
	Electronic referrals and specialist letters	Yes	6 (27%)
	Health care provider/service directories	Yes	8 (36%)
	Alerts monitoring and management	Yes	8 (36%)
Health care service delivery tools	Chronic disease management	Yes	5 (23%)
	Clinical decision support	Yes	5 (23%)
	Medications management	Yes	9 (41%)
	Prescription decision support	Yes	7 (32%)
	Real-time clinical data analysis	Yes	4 (18%)
	Telemedicine and m-health	Yes	12 (55%)
	Consumer health knowledge sources	Yes	4 (18%)
Health information and knowledge	Distance learning/electronic resources	Yes	9 (41%)
	Health care provider knowledge sources	Yes	8 (36%)
	Adverse event monitoring	Yes	8 (36%)
Health care management and administration	Clinical practice improvement	Yes	4 (18%)
	Compliance monitoring	Yes	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	Yes	5 (23%)
	Risk analysis	Yes	3 (14%)
	Surveillance	Yes	9 (41%)

Sudan

Country context			
World Bank: country population (millions)	46.874	World Bank: area (km²)	1 886 68
World Bank: income group	Low income	ITU: individuals using the Internet (2020)	28.40%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	Yes	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	Yes	9 (40%)
National e-health (or digital health) policy or strategy	Yes	7 (31%)
National health information system (HIS) policy or strategy	Yes	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	Yes	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	No	9 (41%)
Common terminology standards	No	12 (55%)
Software accreditation standards	Yes	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	No	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	No	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	No	8 (36%)
	Alerts monitoring and management	No	8 (36%)
Health care service delivery tools	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	No	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	No	12 (55%)
	Consumer health knowledge sources	No	4 (18%)
Health information and knowledge	Distance learning/electronic resources	No	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
	Adverse event monitoring	No	8 (36%)
Health care management and administration	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	No	9 (41%)

Syrian Arab Republic

Country context			
World Bank: country population (millions)	22.125	World Bank: area (km ²)	185 180
World Bank: income group	Low income	ITU: individuals using the Internet (2020)	35.78%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	No	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	No	9 (40%)
National e-health (or digital health) policy or strategy	No	7 (31%)
National health information system (HIS) policy or strategy	In process	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	In process	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	No	9 (41%)
Common terminology standards	Yes	12 (55%)
Software accreditation standards	Yes	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	Yes	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	Yes	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	Yes	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	No	8 (36%)
Health care service delivery tools	Alerts monitoring and management	No	8 (36%)
	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	Yes	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	Yes	12 (55%)
Health information and knowledge	Consumer health knowledge sources	No	4 (18%)
	Distance learning/electronic resources	No	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
Health care management and administration	Adverse event monitoring	No	8 (36%)
	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	Yes	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	No	9 (41%)

Tunisia

Country context			
World Bank: country population (millions)	12.356	World Bank: area (km ²)	163 610
World Bank: income group	Lower middle income	ITU: individuals using the Internet (2020)	71.90%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	In process	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	Yes	9 (40%)
National e-health (or digital health) policy or strategy	No	7 (31%)
National health information system (HIS) policy or strategy	No	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	In process	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Yes	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	No	9 (41%)
Common terminology standards	Yes	12 (55%)
Software accreditation standards	No	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	Yes	12 (55%)
	Personal health records	Yes	3 (14%)
	Electronic health records	Yes	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	Yes	9 (41%)
	Appointment booking and management	Yes	6 (27%)
	Care plan management	Yes	3 (14%)
	Electronic health event summaries	Yes	7 (32%)
	Electronic referrals and specialist letters	Yes	6 (27%)
	Health care provider/service directories	Yes	8 (36%)
	Alerts monitoring and management	Yes	8 (36%)
Health care service delivery tools	Chronic disease management	Yes	5 (23%)
	Clinical decision support	Yes	5 (23%)
	Medications management	Yes	9 (41%)
	Prescription decision support	Yes	7 (32%)
	Real-time clinical data analysis	Yes	4 (18%)
	Telemedicine and m-health	Yes	12 (55%)
	Consumer health knowledge sources	Yes	4 (18%)
Health information and knowledge	Distance learning/electronic resources	Yes	9 (41%)
	Health care provider knowledge sources	Yes	8 (36%)
	Adverse event monitoring	Yes	8 (36%)
Health care management and administration	Clinical practice improvement	Yes	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	Yes	4 (18%)
	Health policy development	Yes	5 (23%)
	Operations management	Yes	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	No	9 (41%)

United Arab Emirates

Country context			
World Bank: country population (millions)	9.441	World Bank: area (km ²)	83 600
World Bank: income group	High income	ITU: individuals using the Internet (2020)	100.00%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	Yes	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	Yes	9 (40%)
National e-health (or digital health) policy or strategy	Yes	7 (31%)
National health information system (HIS) policy or strategy	In process	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	Yes	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	In process	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	Yes	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	Yes	9 (41%)
Common terminology standards	Yes	12 (55%)
Software accreditation standards	No	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	Yes	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records)	Yes	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	Yes	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	Yes	7 (32%)
	Electronic referrals and specialist letters	Yes	6 (27%)
	Health care provider/service directories	Yes	8 (36%)
	Alerts monitoring and management	Yes	8 (36%)
Health care service delivery tools	Chronic disease management	No	5 (23%)
	Clinical decision support	Yes	5 (23%)
	Medications management	Yes	9 (41%)
	Prescription decision support	Yes	7 (32%)
	Real-time clinical data analysis	No	4 (18)%
	Telemedicine and m-health	No	12 (55%)
	Consumer health knowledge sources	No	4 (18%)
Health information and knowledge	Distance learning/electronic resources	Yes	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
	Adverse event monitoring	Yes	8 (36%)
Health care management and administration	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	Yes	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	Yes	3 (14%)
	Surveillance	Yes	9 (41%)

Yemen

Country context			
World Bank: country population (millions)	33.697	World Bank: area (km²)	527 968
World Bank: income group	Low income	ITU: individuals using the Internet (2017)	26.71%

Digital transformation commitment and foundation	Country response	Regional yes
Recognizing ICT in health-related strategies and policies		
National health policy or strategy clearly refers to using ICT for health	No	14 (63%)
National universal health coverage policy or strategy refers to using ICT for health	No	9 (40%)
National e-health (or digital health) policy or strategy	No	7 (31%)
National health information system (HIS) policy or strategy	Yes	12 (54%)
Strategies to establish electronic health information exchange (ICT infrastructure and core services)	No	8 (36%)
Recognizing legal aspects		
Legislation(s) to protect the privacy of personally identifiable data of individuals	Do not know	14 (63%)
Developing workforce capacity on digital tools		
Country provides in-service training for health professionals on ICT for health	No	9 (40%)
Recognizing standardization and interoperability		
Data structure standards	No	9 (41%)
Common terminology standards	No	12 (55%)
Software accreditation standards	No	7 (32%)

Existing digital health applications			
Individual electronic health information	Electronic medical records	No	12 (55%)
	Personal health records	No	3 (14%)
	Electronic health records	No	6 (27%)
Health care communication and collaboration	Access to electronic health records and test results	No	9 (41%)
	Appointment booking and management	No	6 (27%)
	Care plan management	No	3 (14%)
	Electronic health event summaries	No	7 (32%)
	Electronic referrals and specialist letters	No	6 (27%)
	Health care provider/service directories	No	8 (36%)
	Alerts monitoring and management	No	8 (36%)
Health care service delivery tools	Chronic disease management	No	5 (23%)
	Clinical decision support	No	5 (23%)
	Medications management	No	9 (41%)
	Prescription decision support	No	7 (32%)
	Real-time clinical data analysis	No	4 (18%)
	Telemedicine and m-health	No	12 (55%)
	Consumer health knowledge sources	No	4 (18%)
Health information and knowledge	Distance learning/electronic resources	No	9 (41%)
	Health care provider knowledge sources	No	8 (36%)
	Adverse event monitoring	No	8 (36%)
Health care management and administration	Clinical practice improvement	No	4 (18%)
	Compliance monitoring	No	4 (18%)
	Health care and clinical research	No	4 (18%)
	Health policy development	No	5 (23%)
	Operations management	No	5 (23%)
	Risk analysis	No	3 (14%)
	Surveillance	No	9 (41%)



Annex 2. **Digital health national strategy cross-check using survey response and publicly available data**

Country	Survey	Publicly available data	Web link
Afghanistan	Yes	No	
Bahrain	Yes	Yes	https://www.moh.gov.bh/Ministry/MOH2030vision
Djibouti	In process	No	
Egypt	Yes	Yes	https://mcit.gov.eg/en/ICT_Strategy
Iran (Islamic Republic of)	Yes	Yes	www.irantelemed.ir/en/NewsDetails.aspx?id=4909
Iraq	In process	No	
Jordan	Yes	Yes	https://petra.gov.jo/Include/InnerPage.jsp?ID=45420&lang=en&name=en_news
Kuwait	In process	In process	
Lebanon	In process	Yes	https://www.moph.gov.lb/en/Pages/6/2651/national-e-health-program
Libya	No	No	
Morocco	Yes	No	
Oman	Yes	In process	
Pakistan	In process	Yes	http://digitalpakistan.pk/pdf/d-health.pdf
Palestine	Yes	No	
Qatar	Yes	Yes	https://www.moph.gov.qa/english/strategies/National-Health-Strategy-2018-2022/Pages/default.aspx
Saudi Arabia	Yes	Yes	https://www.moh.gov.sa/en/Ministry/nehs/Pages/default.aspx
Somalia	In process	In process	
Sudan	Yes	Yes	
Syrian Arab Republic	In process	No	
Tunisia	No	No	
United Arab Emirates	In process	Yes	https://www.doh.gov.ae/-/media/B5F0E47029F04CD8AD4CA16DC1FDBC66.ashx
Yemen	Yes	No	



Annex 3.

Electronic health records cross-check using survey response and publicly available data

Country	Survey	Publicly available data	Web link
Afghanistan	No	No	
Bahrain	Yes	Yes	https://www.moh.gov.bh/Services/MedicalRecords?lang=en
Djibouti	No	No	
Egypt	No	No	
Iran (Islamic Republic of)	Yes	No	
Iraq	No	No	
Jordan	No	In process	
Kuwait	No	In process	
Lebanon	No	In process	https://www.moph.gov.lb/userfiles/files/Programs%26Projects/NationalE-healthProgram/E-Health%20Solutions-2014.pdf
Libya	No	No	
Morocco	No	No	
Oman	Yes	Yes	
Pakistan	No	In process	
Palestine	No	No	
Qatar	No	Yes	
Saudi Arabia	Yes	Yes	https://www.moh.gov.sa/en/Ministry/vro/EHealth/Documents/MoH-Digital-Health-Strategy-Update-2018.pdf
Somalia	No	No	
Sudan	Yes	In process	
Syrian Arab Republic	No	No	
Tunisia	No	No	
United Arab Emirates	Yes	Yes	https://mohap.gov.ae/en/media-center/news/27/1/2022/mohap-aws-ink-mou-on-cloud-based-electronic-medical-record-system
Yemen	No	No	



Annex 4. **Reported national health priorities**

Country	What are the top national health priorities in your country (3–5 priorities)?
Afghanistan	Armed conflict injuries Communicable diseases Mother and child morbidities and mortalities
Bahrain	Adopting an integrated and sustainable health care approach with attention to health promotion Quality and safety of health services A strong and effective infrastructure for the operation of health information systems and e-health
Egypt	Standardization of Egyptian core health indicators Building capacity and training of health informatics specialists Using health informatics standards in coding vital records (ICD-10 for the cause of death) “Egyptian Knowledge Bank”: this is one of the largest national educational projects in Egypt. It aims to provide a huge and diversified source of knowledge and culture for free, to all Egyptians Automation and shifting the focus on digital transformation plus the introduction and use of International Classification for Diseases 11th Revision (ICD-11) and International Classification of Health Interventions (ICHI) Preparedness and management of COVID-19 epidemic Presidential public health initiatives Universal health insurance programme
Iran (Islamic Republic of)	COVID-19-related priorities New variants of COVID-19 Access to COVID-19 vaccines for high-risk and priority age groups Continuity of essential health services Electronic birth and death certificates as part of civil registration and vital statistics system Strengthening health service provision based on family practice in urban areas and using digital referral system
Iraq	Improving emergency preparedness and response systems Improving health financing mechanisms and expanding funding for health Regulating the pharmaceutical sector and strengthening the regulatory functions of medicines, including drug selection, registration, monitoring, post-marketing follow-up and updating of drug supply, storage and distribution policies and systems Health workforce planning, addressing gaps in the health workforce, especially maldistribution and developing a clear plan of action for human resources Strengthening family practice-based health services Strengthening the health information system
Jordan	Improving effectiveness and efficiency of knowledge management based on digital transformation and technology Providing equitable and high-quality health care services Improving effectiveness and efficiency of infrastructure management Improving effectiveness and efficiency by improving human resources management Maximizing governance and the supervisory role of the Ministry and implementing decentralization Improving effectiveness and efficiency of financial resources management Ministry of Health Strategic Plan 2018–2022 Increasing inclusion of citizens in universal health coverage
Kuwait	Noncommunicable diseases, public health and emergency response, primary care, digital health
Lebanon	Noncommunicable diseases, EWARS, maternal and child health
Libya	Lack of medicines, lack of skilled human resources, hard-to-reach inaccessible areas

(continued)

Country	What are the top national health priorities in your country (3–5 priorities)?
Morocco	Extending health coverage for all the resident population in Morocco
	Human resources development: revising the law on the practice of medicine; creation of a public health service
	Health governance: establishing regional health agencies and territorial health groups
	Upgrading and strengthening of health care supply: development of primary health care and hospital supply
	Developing an integrated health information system
	Strengthening core public health functions and health security
Oman	Noncommunicable diseases
	Financing
	Emerging and communicable diseases
	Emergency preparedness and response
	Women, child and geriatric health
Pakistan	Nutrition, population, tuberculosis, Expanded Programme on Immunization and noncommunicable diseases
Palestine	Ensuring the provision of comprehensive health care services for all citizens by extending health services nationally
	Promoting programmes for the management of communicable diseases and noncommunicable diseases, preventive health care and community health awareness
	Institutionalizing quality systems in all aspects of health service delivery
	Providing adequate, efficient and qualified human resources in the Palestinian health sector
	Enhancing health governance and institutional development, including enforcement of laws and legislation, cross-sectoral coordination and integration among service providers, enhancing health financing and financial protection against health costs
Qatar	Health and well-being for people with special needs
	Healthy children and adolescents
	Healthy women leading to healthy pregnancy
	Improving health for people with chronic conditions
	Improving mental health and well-being
Saudi Arabia	Health care digital transformation strategy throughout national e-health and sustainable financing of the health sector
	Improving national health workforce development and skills
	Improving health and quality of life for the public by protection and prevention; main burden of disease is NCDs
	Maintaining the safety of mass gatherings (during hajj and umrah)
	Managing pandemics and crises in health care cross the country, such as COVID-19
Somalia	Health system strengthening; maternal neonatal and child health; emergency preparedness and response; control of communicable diseases
Sudan	Increasing the reporting rate of health facilities through the digital health information systems (90%)
	Hospital management information system
	Surveillance system and integration
	E-services
Syrian Arab Republic	Building national team capacity in health information systems
	Supporting the Ministry of Health and Ministry of Higher Education and Scientific Education with electronic medical record system in all public hospitals
	Supporting the country with licences for important e-health services and programs e.g. MS Windows, SQL, GIS, SQL Server Reporting Services
Tunisia	Developing a package of essential services for everybody

(continued)

Country	What are the top national health priorities in your country (3–5 priorities)?
	Ensuring a unified basic benefit package for everybody
	Making the health care system revolve around family and local health
	Protecting the health of citizens
	Providing necessary means to the public sector to guarantee its service missions
United Arab Emirates	Finalizing unified medical record system; activating e-health policies and regulations; population health
Yemen	Communicable disease control, elimination, eradication
	Reducing mortality and morbidity related to maternal and child health
	Tackling malnutrition



Annex 5. **WHO regional survey tool**

Q #	Survey section	Question
Q1	(I) Basic information, foundation of digital health	What are the top national health priorities in your country (3–5 priorities)?
Q2		Does your country have a national health policy or strategy that clearly refers to the use of ICT or e-health?
Q3		Does your country have a national universal health coverage policy or strategy that clearly refers to the use of ICT or e-health?
Q4		Does your country have a national e-health (or digital health) policy or strategy?
Q5		Does your country have a national health information system (HIS) policy or strategy?
Q6		Does your country have strategies to establish electronic health information exchange (ICT infrastructure and core services)?
Q7		Does your country have legislation(s) to protect the privacy of personally identifiable data of individuals, irrespective of whether it is in paper or digital format?
Q8		Does your country provide in-service training for capacity-building of health professionals in the use of ICT for health?
Q9	(II) e-health services and application components	Individual electronic health information
Q10		Health care communication and collaboration
Q11		Health care service delivery tools
Q12		Health information and knowledge
Q13		Health care management and administration
Q14		Others
Q15	(III) e-health standards and interoperability components	Data structure standards: These standards govern the way health datasets are stored using consistent data structures and can be presented with consistency in software applications, to ensure that information is neither misinterpreted nor overlooked
Q16		Common terminologies: These enable information communicated electronically to make use of a common language for describing symptoms, diagnoses and treatments, e.g. clinical coding standards or medicines terminology standards.
Q17		Software accreditation standards: These define the criteria with which digital health software products and services must comply in order to be certified as being able to exchange health information with the national digital health environment
Q18		Others
Q19	(IV) e-health national governance and guidance (including focal points information)	Entity name
Q20		Entity URL
Q21		Entity category
Q22		Entity contact person name (focal point)
Q23		Focal point position
Q24		Focal point email
Q25		Second entity name
Q26		Second entity URL
Q27		Second entity category
Q28		Second entity contact person name (focal point)
Q29		Second focal point position
Q30		Second focal point email
Q31		Third entity name
Q32		Third entity URL
Q33		Third entity category
Q34		Third entity contact person name (focal point)
Q35		Third focal point position
Q36		Third focal point email

(continued)

Q #	Survey section	Question
Q37	(V) Perceived barriers to adopting and implementing e-health programmes	Capacity; lack of trained human resources and/or technical support
Q38		Infrastructure; lack of equipment and/or connectivity
Q39		Funding; lack of funding to develop and support
Q40		Effectiveness; lack of evidence on effectiveness of programmes
Q41		Cost-effectiveness; lack of evidence on cost-effectiveness of programmes
Q42		Demand; lack of demand for digital health programmes by health professionals or target groups
Q43		Legal; lack of legislation or regulations covering digital health programmes
Q44		Policy; national policies do not recognize digital health in health services delivery
Q45		Priorities; competing health system priorities
Q46		Please describe any other high-impact barriers not mentioned above
Q47	(VI) Digital response to COVID-19 pandemic	Has your country implemented any digital solution(s) for pandemic control
Q48		Please provide web links for easy reference of implemented digital solutions if possible
Q49		Kindly identify major national challenges relevant to implementation of innovative pandemic control solutions
Q50	Final comments	Kindly indicate any further ideas or comments supporting your responses mentioned above

